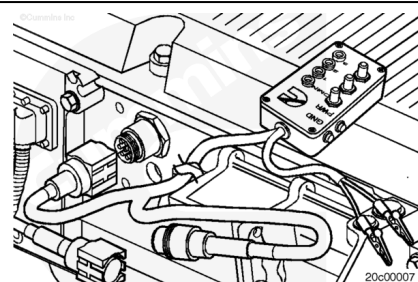


Trainings ▾

Resistance Check

⚠ CAUTION ⚠

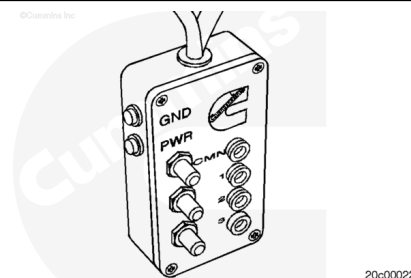
Do not connect the engine breakout cable to the engine wiring harness connector. Connecting the engine brake breakout cable to the engine wiring harness will give a false reading when checking the resistance.



Disconnect the engine wiring harness connector from the cylinder head pass-through connector.

Connect one end of the engine brake breakout cable, Part Number 3163150, to the cylinder pass-through connector.

Note : The black test port on the engine brake breakout cable box is the common ground for all three solenoids. The white test ports (1, 2, and 3) are used to test the resistance of the corresponding brake solenoids (1, 2, and 3).

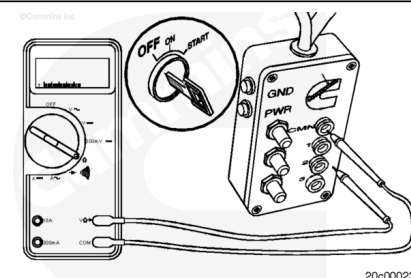


Note : The ignition key must be in the OFF position.

Check the resistance of each solenoid.

- **Cold engine** - 8 to 10 ohms
- **Hot engine** - 16 to 20 ohms.
- **Cold engine** - 32 to 40 ohms
- **Hot engine** - 60 to 80 ohms.

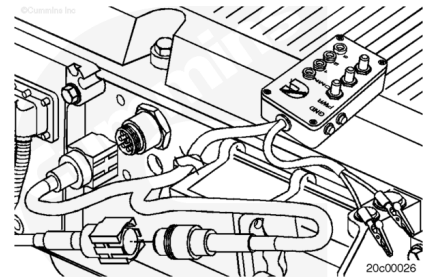
Note : If not within specification, using the engine brake breakout tool, the engine brake wiring harness and solenoids must be checked separately to identify the faulty component. Refer to Procedure 020-012 (Engine Brake Solenoid Valve) in Section 20. (</qs3/pubsys2/xml/en/procedures/10/10-020-012.html>) Refer to Procedure 020-015 (Engine Brake Wiring Harness) in Section 20.



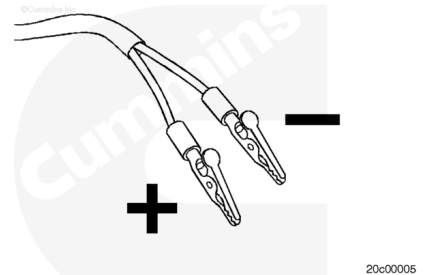
Voltage Check

Disconnect the engine wiring harness connector from the cylinder head pass-through connector.

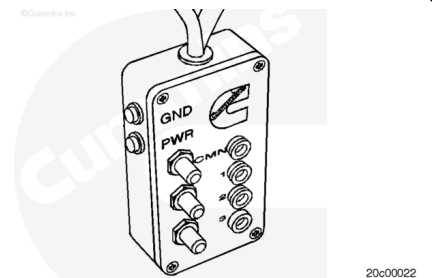
Connect one end of the engine brake breakout cable, Part Number 3163150, to the cylinder pass-through connector and the other end to the engine wiring harness connector.



Connect the red alligator clip of the engine brake breakout cable to battery voltage and the black alligator clip to ground.



Note : The black test port on the engine brake breakout cable box is the common ground for all three solenoids. The white test ports (1, 2, and 3) are used to read the voltage of the corresponding brake solenoids 1, 2, and 3.

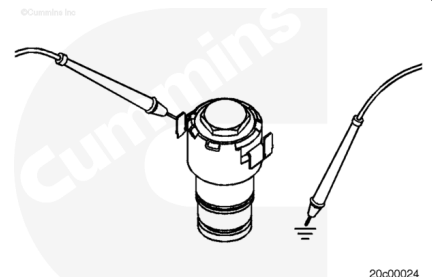


Note : The ignition key must be in the ON position.

Check the voltage of each solenoid.

The voltage should be 12-VDC for each solenoid.

Note : If not within specification, using the engine brake breakout tool, the engine brake wiring harness and solenoids must be checked separately to identify the faulty component. Refer to Procedure 020-012 (Engine Brake Solenoid Valve) in Section 20. (/qs3/pubsys2/xml/en/procedures/10/10-020-012.html) Refer to Procedure 020-015 (Engine Brake Wiring Harness) in Section 20.

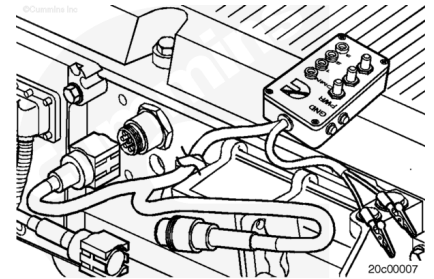


Initial Check

Engine Brake Solenoid

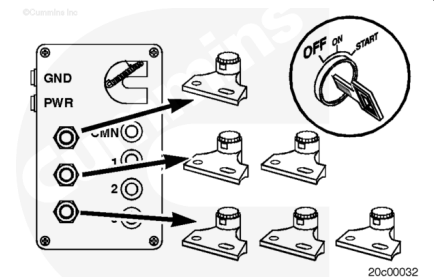
Disconnect the engine wiring harness connector from the cylinder head pass-through connector.

Connect one end of the engine brake breakout cable, Part Number 3163150, to the cylinder pass-through connector and the other end to the engine wiring harness connector.



Note : The ignition key must be in the OFF position.

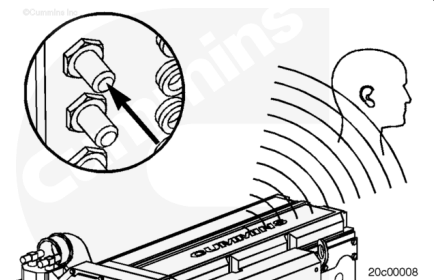
- Button No. 1 actuates solenoid No. 1 and brake Number 1.
- Button No. 2 actuates solenoid No. 2 and brakes Number 2 and 3.
- Button No. 3 actuates solenoid No. 3 and brakes Number 4, 5, and 6.



Press the engine brake breakout cable box button (1, 2, or 3) to activate the corresponding engine brake solenoid (1, 2, or 3).

Listen for a clicking sound when activating the brake solenoid. If no clicking sound can be heard when the button is pressed, the corresponding solenoid valve and the engine brake wiring harness **must** be checked separately to identify the faulty component. Refer to Procedure 020-012 (Engine Brake Solenoid Valve) in Section 20.

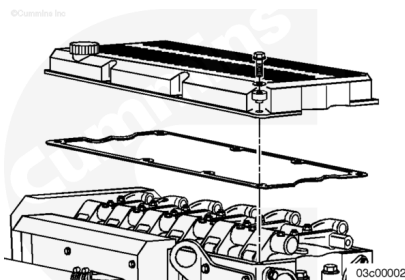
(/qs3/pubsys2/xml/en/procedures/10/10-020-012.html) Refer to Procedure 020-015 (Engine Brake Wiring Harness) in Section 20. (/qs3/pubsys2/xml/en/procedures/10/10-020-015.html)



Engine Brake Rocker Lever

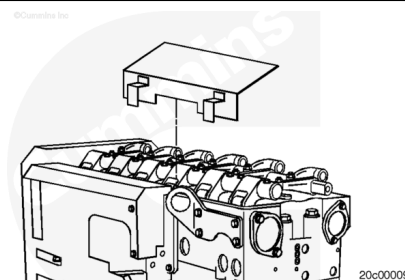
Remove the rocker lever cover. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3.

(/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html)



Install the splash shield , Part Number 3163092, to reduce the amount of oil being thrown by the fuel pump drive gear.

Start the engine.



Press the engine brake breakout cable box button (1, 2, or 3) to activate the corresponding engine brake solenoid (1, 2, or 3).

Check the corresponding engine brake lever(s) and solenoid for the following:

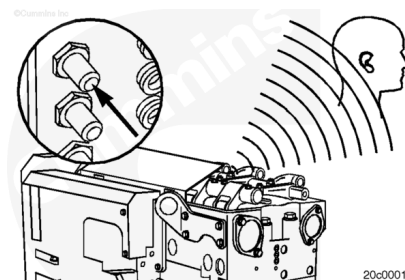
- Engine brake piston deployment
- Leakage from upper seals of solenoid.

Replace parts as needed. Refer to Procedure 020-012 (Engine Brake Solenoid Valve).

(/qs3/pubsys2/xml/en/procedures/10/10-020-012.html) Refer to Procedure 020-017 (Engine Brake Control Valve).

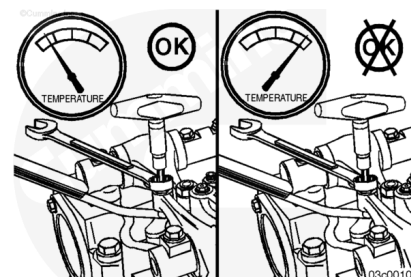
(/qs3/pubsys2/xml/en/procedures/10/10-020-017.html) Refer to Procedure 020-019 (Engine Brake Piston) in Section 20.

(/qs3/pubsys2/xml/en/procedures/10/10-020-019.html)



Adjust

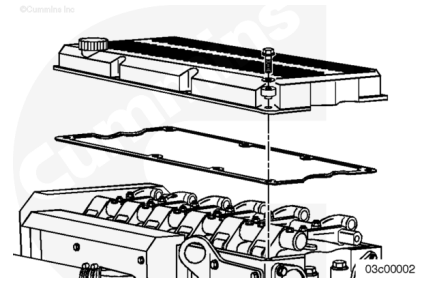
All overhead adjustments **must** be made when the engine is cold (any stabilized coolant temperature at 60°C [140°F] or below).



⚠ CAUTION ⚠

Do not use solvent to clean the rocker cover gasket. Solvent can damage the gasket material and cause it to swell.

Remove the rocker lever cover and gasket. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3.
(/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html)



Note : Early engine vibration dampers are marked with BRAKE SET 1-6, BRAKE SET 2-5, or BRAKE SET 3-4. The engine brakes must be set at the appropriate mark on these engines. Newer engine vibration dampers are marked with only A, B, or C, and are adjusted with the valves and injector on the same cylinder.

Locate the valve set marks on the outside of the vibration damper.

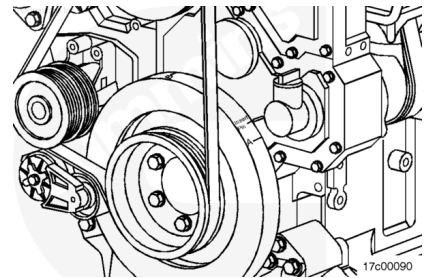
The set marks are A, B, and C:

Set to mark A to adjust cylinder 1 or 6.

Set to mark B to adjust cylinder 2 or 5.

Set to mark C to adjust cylinder 3 or 4.

Note : Two complete revolutions are required to set all valves, engine brakes, and injectors.

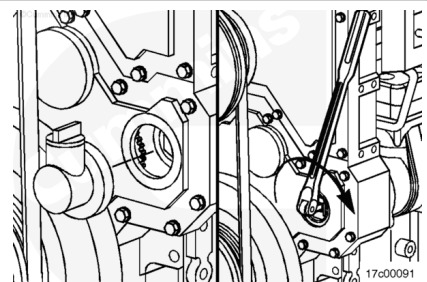


With Air Compressor:

Remove the oil fill connector from the lower gear case cover.

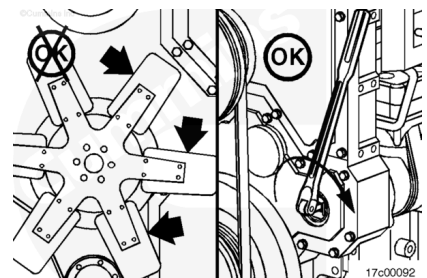
Insert a 3/4-inch drive ratchet and extension into the air compressor drive.

Rotate the air compressor drive **clockwise**, as viewed from the front of the engine.



⚠ WARNING ⚠

Do not pull or pry on the fan to manually rotate the engine. To do so can damage the fan blades. Damaged fan blades can cause premature fan failures which can result in serious personal injury or property damage.



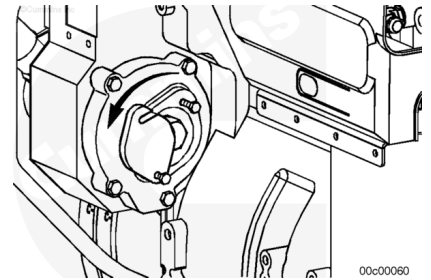
The crankshaft rotation is **clockwise**, as viewed from the front of the engine.

The cylinders are numbered from the front of the engine (1-2-3-4-5-6).

The engine firing order is 1-5-3-6-2-4.

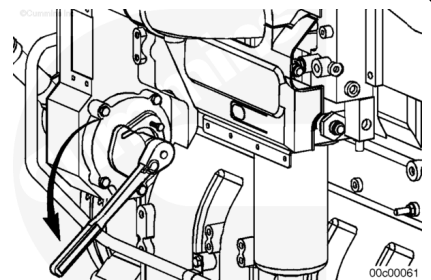
Without Air Compressor:

Loosen the capscrews and rotate the cover or remove the oil fill tube, if equipped.



Use a 1 1/2-inch socket, push the barring gear into the gear mesh and bar engine over **counterclockwise**.

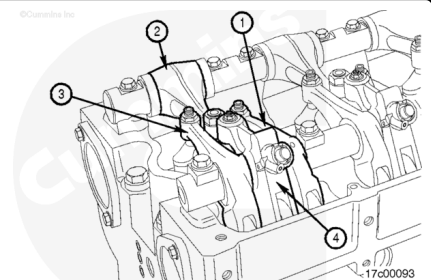
Note : Rock the barring device back and forth until it disengages.



Each cylinder has four rocker levers:

- The exhaust valve rocker lever (1)
- The injector rocker lever (2)
- The intake valve rocker lever (3)
- The engine brake rocker lever (4).

Note : The intake valve rocker lever is always the long lever on the valve rocker lever shaft.



Note : Early engine vibration dampers are marked with BRAKE SET 1-6, BRAKE SET 2-5, or BRAKE SET 3-4.

The engine brakes must be set at the appropriate mark on these engines. Newer engine vibration dampers are marked with only A, B, or C, and are adjusted with the valves and injector on the same cylinder.

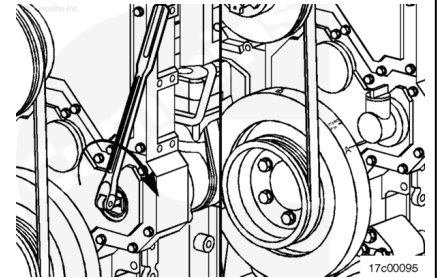
The valves, brakes, and the injectors on the same cylinder are adjusted at the same index mark on the vibration damper.

Signature, ISX and
QSX15

Brake Adjustment Sequence		
Bar Engine in Direction of Rotation	Pulley Position	Brake
Start	A	1
Advance to	B	5
Advance to	C	3
Advance to	A	6
Advance to	B	2
Advance to	C	4
Firing Order: 1-5-3-6-2-4		

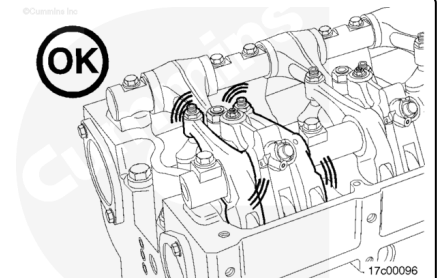
Rotate the compressor drive or barring device in the direction of engine rotation, **clockwise**. Align the A mark on the vibration damper with the pointer on the gear cover.

Note : For illustrative purposes, position A is shown as the first step. It is **not** necessary to start with position A, as long as the proper sequence is followed.



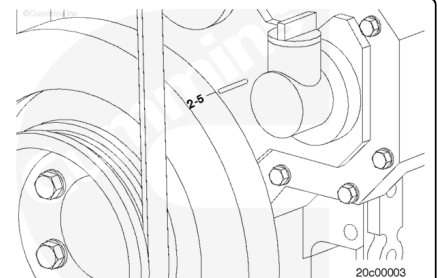
Check the valve rocker levers on the given cylinder to see if both intake and exhaust valves are closed.

Note : Both sets of valves are closed when the rocker levers and the brake lever are loose. If both sets of valves are **not** closed, rotate the compressor drive gear one complete revolution, and align the A mark on the front damper with the pointer again.



⚠ CAUTION ⚠

To get maximum brake operating efficiency and to prevent engine damage, the brake adjustment instructions must be followed.



For older engines, locate the engine brake set marks on the outside of the vibration damper.

The set marks are BRAKE SET 1-6, and BRAKE SET 2-5, and BRAKE SET 3-4.

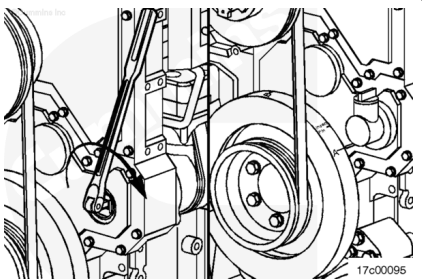
BRAKE SET 1-6: Cylinder 1 or 6 adjust

BRAKE SET 2-5: Cylinder 2 or 5 adjust

BRAKE SET 3-4: Cylinder 3 or 4 adjust

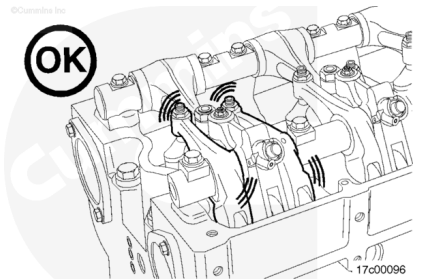
Rotate the compressor drive or barring device in the direction of engine rotation, **clockwise**. Align the A mark on the vibration damper with the pointer on the gear cover.

Note : For illustrative purposes, position A is shown as the first step. It is **not necessary** to start with position A, as long as the proper sequence is followed.

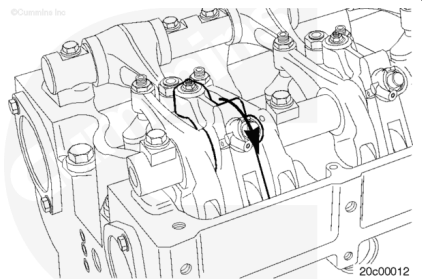


Check the valve rocker levers on the given cylinder to see if both intake and exhaust valves are closed.

Note : Both sets of valves are closed when the rocker levers and the brake lever are loose. If both sets of valves are **not** closed, rotate the compressor drive gear one complete revolution, and align the A mark on the front damper with the pointer again.

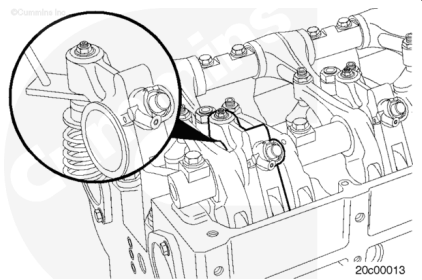


Press the engine brake lever down to verify that the camshaft follower is in contact with the camshaft.



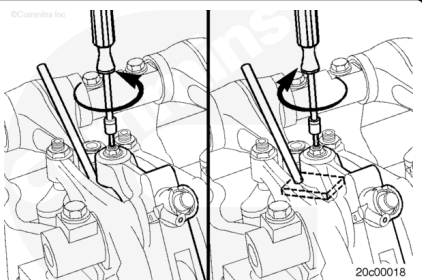
Loosen the locknut on the brake lever adjusting screw, and back out the adjusting screw one turn.

Insert the feeler gauge, Part Number 3163530, between the bottom of the engine brake piston and the top of exhaust valve pin on the exhaust valve crosshead.



mm		in
7.00	NOM	0.276

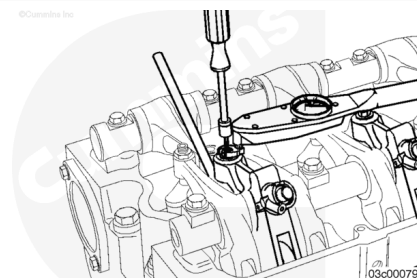
Tighten the adjusting screw until drag on the feeler gauge is felt. Proper drag means that there is no motion of the brake lever camshaft follower against the cam lobe.



Hold the engine brake lever adjusting screw, and tighten the locknut.

Torque Value: 20 n•m [177 in-lb]

Remove the feeler gauge.



⚠ CAUTION ⚠

Engine damage can occur if running clearance is not within specifications.

Check the running clearance:

1. Rotate the engine brake rocker lever to the detent (neutral) position.
2. Check the clearance (1) between the engine brake lever actuator piston and the crosshead guide pin.

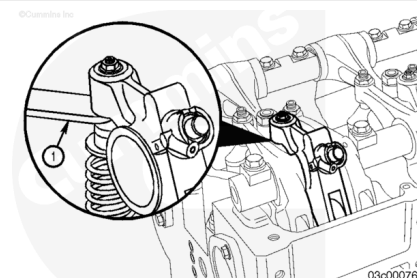
mm		in
0.635	MIN	0.025
2.790	MAX	0.110

If the running clearance does **not** fall in the given specifications, loosen the rocker lever shaft and rotate the rocker lever shaft, slightly, in the direction required to bring the running clearance within specifications.

Recheck the brake running clearance.

Note : The rocker lever shafts must be adjusted so that all three engine brake levers fall within the given running clearance specification.

Repeat the process to adjust all engine brakes according to the chart shown earlier in this procedure.



Install the rocker lever cover and gasket. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3.

(/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html)

If failure resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 (Aftertreatment Testing) in Section 14.

(/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

